

EXECUTIVE ENGINEERING DOCUMENTATION

RDX Request Management & Tracking Utility

A unified corporate solution governing sample allocations, free-of-cost items, 4-tier digital approvals, real-time departmental budgets, and end-to-end warehouse shipment fulfillment.

GOVERNANCE CHAIN

4 Stages

Exec → Mgr → HOD → President

BUDGET PRECISION

100% Real-Time

Auto-deductions & balance locks

LOGISTICS TRACKING

4 Kanban States

Ready → In Process → Dispatched
→ Delivered

USER EXPERIENCE

0ms Latency

Optimistic local cache + cloud sync

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What is the Request Tracking Utility?

Plain-English summary of the business problem and how this software solves it

Core Purpose: Companies lose thousands of dollars each month when team members give away free product samples, client gifts, or promotional items without budget checks or manager approvals. **RDX Request Tracking Utility** automates this entire lifecycle so that no sample leaves the warehouse without authorized signatures and recorded budget deductions.

Before this system, requests were handled through scattered emails, paper slips, or spreadsheets. This created 3 major bottlenecks:

✗ Problem 1: Budget Leakage

Teams gave away samples without knowing if their department still had budget left, causing unplanned deficits.

✗ Problem 2: Lost Approvals

Requests got stuck in executive inboxes for weeks with zero tracking or visibility into who was holding them up.

✗ Problem 3: Warehouse Blindspot

Once approved, no one knew if warehouse staff actually packed, dispatched, or delivered the items to the client.

✓ The RDX Solution:

A single connected dashboard where **Sales/Staff create requests** → **Executives digitally sign** in a strict 4-step sequence → **Budgets auto-deduct** → **Warehouse staff fulfill shipments** on a real-time Kanban board → **Every detail is audited with timestamps and IP stamps**.

1. Request Draft
SKUs & Pricing

2. 4-Stage Approval
Digital Signatures

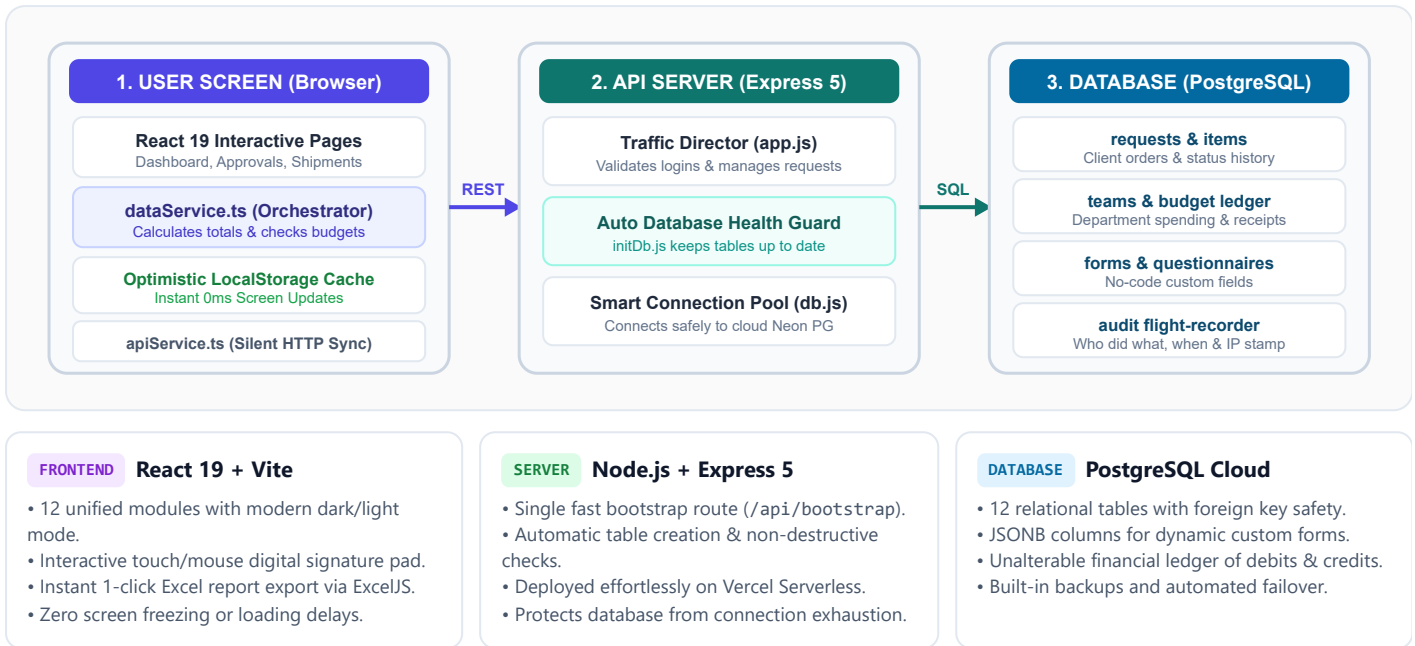
3. Auto Deduction
Balance Ledger

4. Warehouse QA
Pack & Dispatch

5. Delivery
Closed • Safe

2 The 3-Tier Architecture in Plain English

How the web browser, application server, and cloud database talk to each other



How Zero-Lag Works: When you click "Approve", the button turns green in **0 milliseconds** because it updates your browser memory first. It then quietly informs the cloud database in the background. Even if your internet flickers, your work is preserved safely.

Step	What the User Does	What the System Does	Speed
1. User Action	Signs name & clicks "Approve"	Captures digital signature stroke vector & timestamp	Instant
2. Budget Guard	Approving the request	Verifies if department still has enough remaining money	< 1 millisecond
3. Screen Update	Sees green success badge	Saves directly into browser cache with zero waiting	0ms (Real-time)
4. Cloud Save	Continues working freely	Sends quiet background update to PostgreSQL database	Background async
5. Ledger Stamp	Audit trail updated	Writes permanent debit receipt with balance before & after	Permanent record

3 The 4-Stage Approval Journey

How a sample request travels through the corporate hierarchy

STAGE 1 Executive

Commercial Check:

Reviews why the sample is needed, client business history, and specific items requested.

STAGE 2 Manager

Team Alignment:

Confirms operational necessity, sales pipeline priority, and required delivery target dates.

STAGE 3 HOD

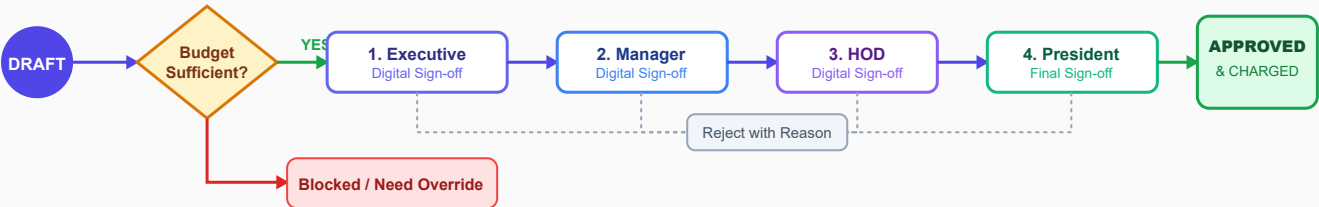
Budget Authority:

Validates that the department has enough remaining money and adheres to gift caps.

STAGE 4 President

Final Sign-Off:

Grants final corporate green light. Auto-deducts team budget and queues shipment.



CALCULATION How Pricing Is Calculated

- **Request Value:** Total retail value of sample items.
- **Formula:** $\text{Budget Amount} = \text{Value} \times (1 - \frac{\text{Discount}}{100})$
- **Hard Stop:** Blocks submission if department money is low.
- **Override Flag:** Super Admins can override, flagged for audit.

EXECUTION Instant Budget Deduction

- **Instant Debit:** When President signs, money is auto-deducted.
- **Receipt Created:** Writes a new row in budget_transactions.
- **Total Clarity:** Records balance_before and balance_after.
- **Zero Disputes:** Links exact order number to the deduction.

Transaction ID	Team / Department	Transaction Type	Deduction	Balance Before	Balance After	Authorized By
TXN-2026-081	Enterprise Sales	REQUEST_DEDUCTION	\$1,250.00	\$48,500.00	\$47,250.00	President (Final Sign-off)
TXN-2026-082	Marketing EMEA	REQUEST_DEDUCTION	\$640.00	\$14,200.00	\$13,560.00	President (Final Sign-off)
TXN-2026-083	Global Accounts	OVERRIDE_DEDUCTION	\$3,100.00	\$1,500.00	-\$1,600.00	Super Admin (Sanctioned)

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Warehouse Fulfillment & Courier Tracking

Real-time Kanban logistics pipeline after request approval

Closing the Loop: Approving a sample is only half the job. The **Shipment Tracking** Kanban board gives warehouse staff and sales agents full visibility into packaging, courier pickup, and confirmed customer delivery.

1. Approved – Ready

President signed off. Order is queued for warehouse stock picking and allocation.

2. In Process

Warehouse team is packing, labeling shipping boxes, and inspecting sample quality.

3. Dispatched

Package handed over to courier. Waybill number and tracking notes are attached.

4. Delivered

Delivery confirmed by client or recipient. System records timestamp and closes request.

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Core Database Tables (In Plain English)

Table Name	What It Stores in Plain English	Why It Matters
requests	Every sample/gift order created, items requested, and client name	Main record tracking the order from draft to delivered
teams	Each corporate department, their total budget, and spent money	Enforces financial spending caps so teams never overspend
budget_transactions	The digital receipt book for every single dollar spent or refunded	Immutable audit proof of balance before and after each approval
users & roles	Employees and who is allowed to sign off at each stage	Ensures only designated managers can sign off at their level
forms & assignments	Custom questionnaires created without writing code	Allows different departments to ask tailored questions on orders
audit_logs	The flight recorder: who did what, when, and from which IP	Complete legal audit trail for internal security and compliance



Zero Budget Waste

Mathematical spending caps prevent unauthorized sample gifts.



5x Faster Approvals

Managers sign securely on phones or laptops in under 30 seconds.



100% Audit Ready

Digital signatures, IP stamps, and balance receipts for every order.



1-Click Excel Reports

Download complete financial and dispatch spreadsheets instantly.

Document Status: Approved • **Classification:** Internal Corporate Specification

Release: Production V1.0 • **Prepared For:** Senior Leadership Review